July 1, 2026

Dear Editor-in-Chief,

We submit the enclosed manuscript entitled "Differential Spectral Damping: Gap-Adaptive Regularization for Ill-Conditioned Kernel Methods" for consideration as a Regular Article in Neurocomputing.

The paper proposes a novel regularization formula for kernel matrix inversion that adapts its penalty to the localized eigengap structure, motivated by the Davis-Kahan eigenvector perturbation theorem. Unlike standard Tikhonov regularization (uniform scalar penalty), DSD selectively suppresses spectrally unreliable eigenvector directions while preserving reliable ones. The method requires no cross-validation or hyperparameter tuning -- parameters are initialized entirely from the spectral structure of the input matrix.

Key results (50-seed paired testing with fairly optimized baselines):

- +4.8pp on real-world GINA (d=970, Cohen d=4.49, p<0.0001)

- +10.4pp at d=200, +2.6pp on Madelon (d=500) over optimally-tuned Tikhonov

- DSD uses zero optimization; Tikhonov receives 15-point grid search

- Validated against Truncated SVD baseline (DSD wins 30/30 seeds)

- Clear operating regime characterized: d>=100, condition number>1000

This work fits Neurocomputing scope in kernel methods, support vector machines, numerical stability, and classification algorithms. The paper includes theoretical motivation (Davis-Kahan based design, algorithmic stability connection) and comprehensive empirical validation on synthetic and real-world datasets. A differentiable PyTorch implementation is provided as open-source software.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. Complete source code and experiment scripts are publicly available at https://github.com/Praveg432/dsd-regularization.

We believe this work will be of interest to Neurocomputing readership in kernel methods, Twin SVM, and numerical methods for machine learning.

Sincerely,

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